

CONSTRUCTION

How to Increase Profit
Margins and Boost
Productivity with Cloud
Field Operations Software

Unearth

Table of Contents

I.	Introduction	1
II.	Declining Construction Productivity and How Software Can Help	2
III.	Effective Field Operations Software Requires a Map	5
IV.	Map-Based Field Operations Increase Transparency and Enable Trust	7
V.	A Quickstart Guide to Buying Construction Field Operations Software	11
VI.	Case Study: Map-Based Software for Dispute Resolution	14

The background of the top section is a dark teal color with a faint, light-colored architectural blueprint or floor plan overlaid. The blueprint shows various rooms, corridors, and structural elements in a technical drawing style.

Introduction

With \$30 trillion in projected spending by 2030, declining project productivity, and low tech adoption - the construction industry is due for a technology overhaul.

From small hand-held devices to enterprise level software, there is a huge range of technology options available to modern construction organizations.

This book will focus on software. We'll explore the issues with past construction project management platforms, how map-based software can help to boost productivity across the board, and how to procure a platform that best addresses your organizations needs.

Declining Construction Productivity and How Technology Can Help

Productivity problems waste time and resources.

McKinsey Global Institute (MGI) found that the construction industry could **boost its value by \$1.6 trillion globally** by bringing its productivity up to speed.

When builds experience consistent delays - due to rework, duplicative processes, and poor communication - the negative financial impact can be significant.

For example - Westinghouse, a nuclear power company, **declared bankruptcy** after continual delays and cost overruns on nuclear power plant projects put them deep in the red.

But what causes poor productivity?



70% [of megaprojects] are
at least 40% late

Effective project management can significantly improve productivity

Review of the research has revealed the following:

“Through our analysis of more than \$1 trillion worth of capital projects over the past five years, we have found that **improving basic project-management skills offers the most potential to improving site performance.**”

- McKinsey Global Institute

“The majority of recommendations [for increasing productivity] involved improving various aspects of project management, including planning, communications, supervision, and personnel and manpower practices.”

- The National Academies Press

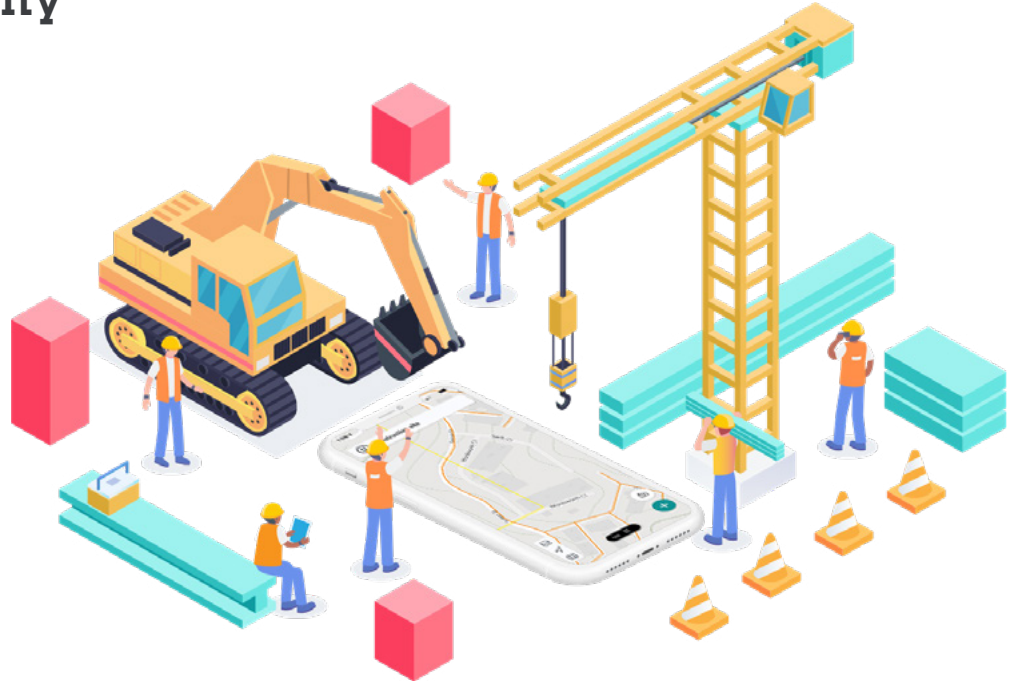


Cloud field operations software helps improve project management and increase productivity

Not all software will solve all problems. Using the wrong software can even create issues.

The right software, however, can make your life easier and your business more effective.

But the right **software must be built with the construction industry's needs in mind.**



Effective Construction Field Operations Software Requires a Map

A construction field operations platform should do three really things well:



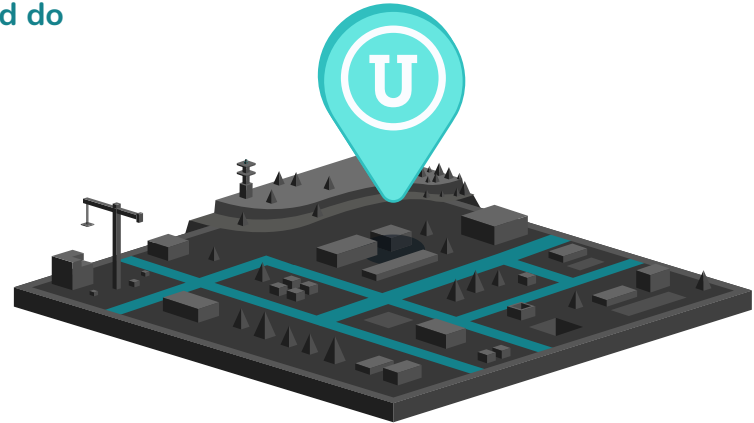
Simplify processes



Document progress



Decrease project duration



Most software falls short of these goals.

Because construction is a physical process tied to a specific location, organizing as you would for other industries - by name, date, or file number - doesn't work.

To be truly effective, construction software must be organized by location using a map-based platform.

Digital maps are the future of construction field operations

Maps are incredibly powerful tools.

Columbus won funding for his voyage to America with a world map. The Waterloo Map helped the British defeat Napoleon, and a series of top secret maps contributed to allied victory on D-Day.

Not only are maps still used in 2019, they are a crucial part of our daily lives. With over one billion monthly users, Google Maps stands as the prime example.

To be genuinely useful for construction projects, **software must reflect the physical world.**



Map-Based Field Operations Increase Transparency and Enable Trust

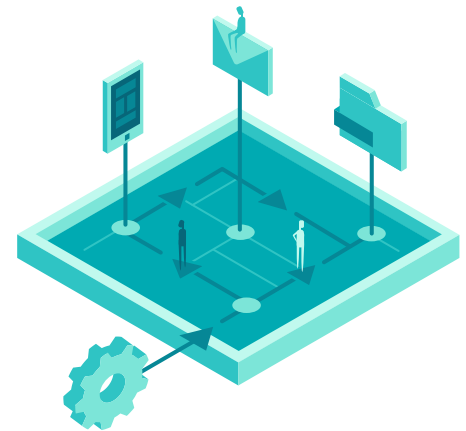
Contractors and owners live in a culture of mistrust - further degrading project productivity.

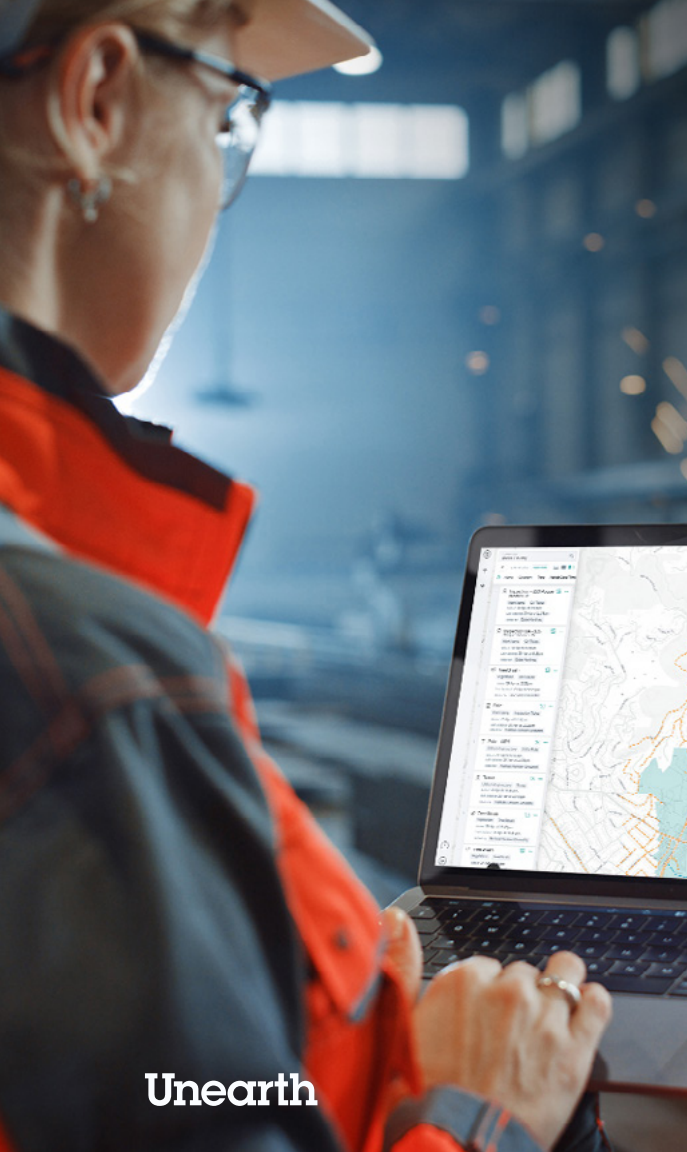
Owners have limited information about what's going on at their build sites, so when a contractor changes the bid - owners are left to wonder if the changes were actually necessary.

For contractors, this means jumping through hoops and documenting everything in order to prove they are operating by the book.

One underlying issue is that owners don't have a simple overview of their projects.

Another is that documents and verbal reports are up to interpretation and thereby easy to dispute.





Create trust by providing easily accessible visual documentation

With an overhead view of the job site, owners can track their projects with ease. With location-based organization, contractors can document work quickly and accurately.

Moreover, location-based visual records break the reliance on written documents. In the event of a dispute, all stakeholders will have access to photographic evidence, accurate measurements, and geo-referenced blueprints.

With less to interpret, there's less to dispute.

Communication, trust, and understanding all increase when everyone has access to the same database of digital data.

Decrease rework with real-time data sharing and communication

A study done by the Mediterranean Journal of Social Sciences reports that **30% of the work done by construction companies is rework.**

Another study reports that **rework can cost up to 5% of total project value.** For projects valued in the millions, even 5% is a significant sum.

Rework has many causes: unclear instructions, slow design, poor documentation, unrealistic timelines, and simple human error to name just a few.

Map-based software solves many of these issues. It speeds up communication, organizes documentation, and provides quick access to vital information.



Process and store data collected in the field by connecting hardware and software



For software and hardware to work together effectively, two things need to be true:

1. Hardware must be reliable, user-friendly, and cost-effective
2. Software must be able to import and organize data captured by hardware

Sensors, drones, mobile-devices, and self-guided machinery produce massive amounts of data. Construction software must be able to collect and display this data effectively.

A Quickstart Guide to Buying Construction Field Operations Software

As you start exploring your software options, here are four questions you should consider:

1. What are your pain points?

Consider the most frustrating, complicated, or expensive part of your workday. Create a list of these issues and use that for reference.

2. What's too expensive?

Construction software allows you to organize and track expenditures, giving you the information you need to expand profit margins and compete effectively.

3. What needs improvement?

Maybe you just want to streamline your operation and expand project capacity. Consider aspects of your business that may be limiting growth.

4. Who needs access?

Many software companies charge per user. If you want to bring on your entire organization, find a platform with an open licensing structure.

6 steps for purchasing field operations software

1. Do your research

Identify your needs and create a spreadsheet listing ideal features. Research your options and take notes on how each product addresses your wishlist.

2. Talk to your team

Don't skip this step! If your team doesn't like a software, they won't use it effectively - no matter what features are offered.

3. Get an online demo

Don't pay for a software until you see it in action. Request demos from each company. Ask them to walk you through, explain relevant features, and create a quote.





4. Do a cost/benefit analysis

By now you should have narrowed your search to two good options. Analyze the cost and estimated savings to see which software will offer the most value.

5. Run it by your team again

Get feedback about the final two options. Remember, if your team doesn't like a software they won't use it to its full capacity - decreasing any potential benefits.

6. Try the software on a test project

Test runs are essential. They allow you to troubleshoot issues in a controlled, low-stakes environment. Roll out your new software on one project. Watch how your team interacts with it and make sure it performs as expected.

Case Study: OnePlace for Dispute Resolution

The problem

The city of El Paso contracted Saab Site to excavate a storm water dam. To avoid hiring a new survey crew, the city took bids based on a survey from the 1970's. While excavating, the contractor soon realized they were moving more dirt than planned and filed a change order.

Work continued and the contractor kept tickets for the number of trucks leaving the site. The city agreed to address the order later. After the project was complete, the city disputed the contractor's change order, accusing him of half filling trucks to increase his own revenue.

The results



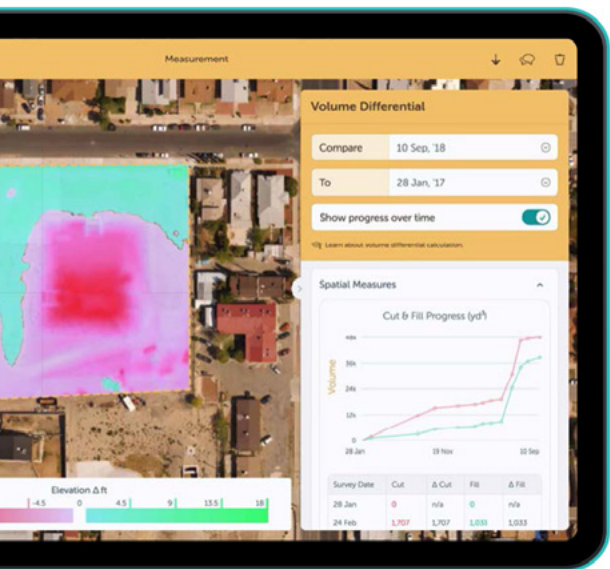
Contractor provided
photographic evidence of
project progression



Contractor received full
payment for services
provided



Agreement was reached and
unnecessary litigation
was avoided



The solution

Having used Uneath's OnePlace™ software to regularly capture aerial site images, the contractor was able to produce detailed progress documentation from every month of the project.

These images showed full trucks leaving the site. They also provided accurate volumetric measurements of the excavated area.

When presented with clear evidence, the city paid the contractor in full and litigation was avoided.

With the documentation provided by OnePlace™, both the contractor and the city avoided months in lost time and saved thousands in attorney fees.



Photo Documentation

Geo-located photos allowed contractor to show an entire visual history of the project



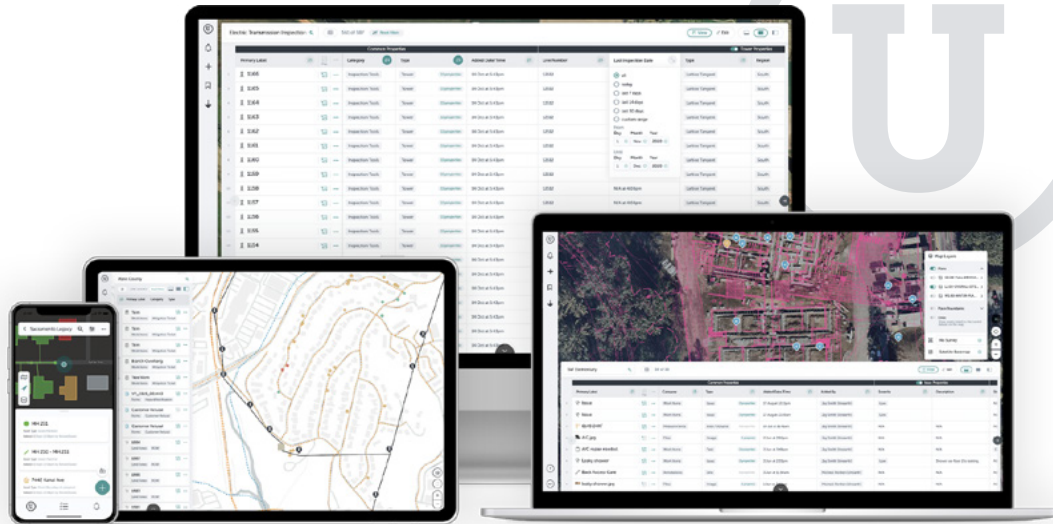
Volumetric Measure

Show volume changes over time using date-stamped drone photography



Drone processing

Continually process and stitch individual drone flights into one complete view



The right software will save you time, energy, and money.
 Harness innovation, increase productivity, improve your bottom line.

[Learn more](#)